GOVERNMENT OF KARNATAKA
DEPARTMENT OF SCHOOL EDUCATION (PRE UNIVERSITY)
LESSON BASED ASSESSMENT I UNIT TEST, JULY-2026
SUBJECT : MATHEMATICS SUBJECT CODE : 35 CLASS : II P U C
MAX MARKS: 30 TIME :1 HOUR

MODEL QUESTION PAPER

Instructions: The question paper has four parts namely A, B, C and D. Answer all the parts.

PART – A

I Answer ALL the questions **8×1=8**

1. A relation R in the set A is called a reflexive relations, if

   (A) (a,a) ∈ R, for every a∈A,

   (B) (a,a) ∈ R, at least one a ∈A

   (C) if (a,b) ∈ R implies that (b, a) ∈ R, for all a, b ∈ A

   (D) if (a,b)and (b,c) ∈ R implies that (a,c) ∈ R, for all a, b,c ∈ A

2. If a relation R on the set {1, 2, 3} be defined by R = {(1, 2)},then R is

   (A) reflexive (B) transitive (C) symmetric (D) equivalence relation.

3. Let f :R → R be defined by $f(x)= \frac{1}{x}$, x ∈R. Then $f$is

   (A) one-one (B) onto (C) bijective (D) $f$is not defined.

4. Let f:$R\rightarrow$ defined by $f(x)$=2x+6 which is a bijective mapping then$f^{-1}(x)$ is given by

   (A) $\frac{x}{2}$ −3 (B) 2x+6 (C) x−3 (D) 6x+2.

5. The domain of f(x)=$sin^{-1}x$.

   (A) (−1,1) (B)[0,π] (C) (−∞,∞) (D) [−1,1].

6. The range of $sec^{-1}xis$

   (A) [-π/2, π/2]−{0} (B)(0,π)−$\frac{\pi}{2}$(C) [0,π]−$\frac{\pi}{2}$ (D) R−(−1,1)

7. Principal value of $cos^{-1}\left(-\frac{1}{2}\right)$ is

   (A)−$\frac{\pi}{6}$ (B) −$\frac{\pi}{3}$(C)$\frac{5\pi}{6}$ (D)$\frac{2\pi}{3}$

8. $If sin^{-1}x = y\,, then$

   (A) $0 \le y \le \pi$ (B) $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$ $(C) 0<y<\pi$ (D) $0<y<\pi$.

II Fill in the blanks by choosing the appropriate answer from those given in the bracket(1,6,8) **2×1=2**

9. The number of all one-one functions from set A = {1, 2, 3} to itself is -----

10. $\sin\left[\frac{\pi}{3}-\sin^{-1}\left(-\frac{1}{2}\right)\right] isequalto$----------

PART B

III. Answer any TWO questions: **2×2=4**

11. Show that $sin^{-1}\left(2x\sqrt{1-x^2}\right) - 2\,tan^{-1}x,\ -\frac{1}{\sqrt{2}} \le x \le \frac{1}{\sqrt{2}}$

12.Find the value of $tan^{-1}\sqrt{3} - sec^{-1}(-2)$

13. Write in the simplest form:$tan^{-1}\left(\frac{x}{\sqrt{a^2-x^2}}\right), |x| < a$

PART C

IV. Answer any TWO questions: **2×3=6**

14. Determine whether the Relation R in the set A = {1,2,3,...,13,14 } defined as $R = \{(x,y): 3x - y = 0\}$ is reflexive, symmetric & transitive:

15. Prove that $sin^{-1}\left(\frac{3}{5}\right) + cos^{-1}\left(\frac{12}{13}\right) = sin^{-1}\left(\frac{56}{65}\right)$

16. Let L be the set of all lines in a plane and R be the relation in L defined as R = {(L_1, L_2) : $L_1$is perpendicular to L_2}. Show that R is symmetric but neither reflexive nor transitive.

PART D

V. Answer any TWO questions: **2×5=10**

17. State whether the function is one-one, onto or bijective. Justify your answer.f : R →R defined by $f(x) = 3 - 4x$.

18. State whether the function is one-one, onto or bijective. Justify your answer.f : R →R defined by $f(x) = 1 + x^2$.

19. Let f : N →Y be a function defined as $f(x) = 4x + 3$, where, Y = {y ∈ N: y = 4x + 3 for some x ∈ N}. Show that f is invertible. Find the inverse.

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